17 **B** $I = \dfrac{P}{4\pi r^2}$, $I = kA^2$

$$kA^2 = \frac{P}{4\pi r^2} \quad \Rightarrow \quad A = \sqrt{\frac{P}{4\pi k}} \times \frac{1}{r}$$

Amplitude is inversely proportional to distance (first graph).
Intensity is proportional to amplitude^2 (second graph).

18 **C** 7.5×10^{-3}